# HQU-MSANet Quantum AI Clinical Diagnostic Report

BraTS 2023 Multi-Modal Quantum Uncertainty-Coupled Attention Network

Automated Neuro-radiology Segmentation & Epistemic Uncertainty Quantification

## 1. PATIENT DEMOGRAPHICS & CLINICAL REGISTRATION

Patient Name: Eleanor Rigby

Patient ID / MRN: PAT-9921

Age / Gender: 52 yrs / Female

Scan Timestamp: 2026-09-18 15:48:14

Referring Doc: Dr. Robert Smith, MD

Computing Device: NVIDIA GeForce RTX 3050 6GB Laptop GPU

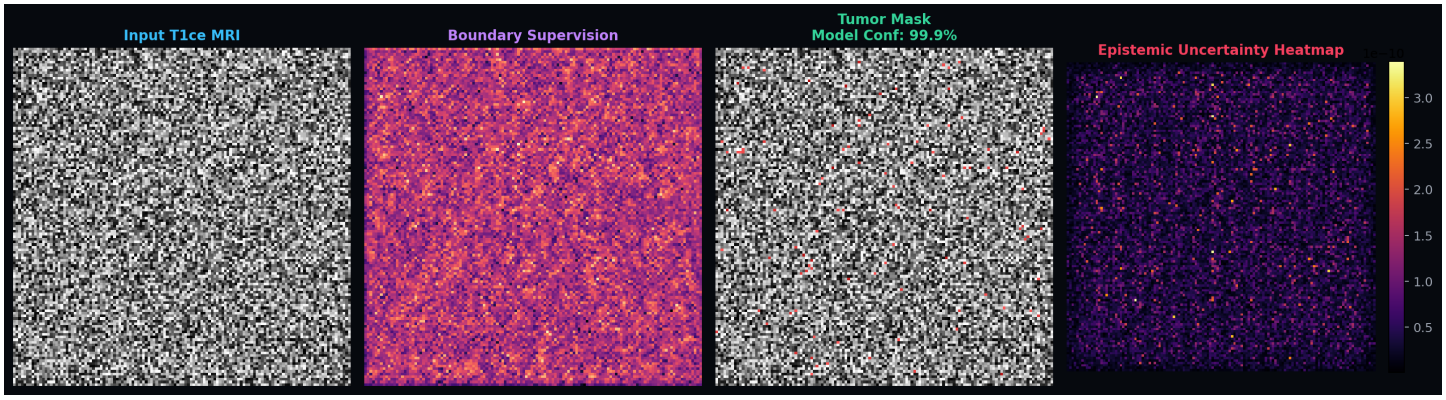
## 2. 7-STAGE PIPELINE BACKEND EXECUTION AUDIT

| Pipeline Stage | Backend Execution Description | Status |
|---|---|---|
| Step 1: Input Scan Ingestion | Multi-modal channels (T1, T1ce, T2, FLAIR) loaded and dimension verified (128x128). | COMPLETED |
| Step 2: Preprocessing & Scaling | Skull-stripping intensity mask applied, Z-score normalized (zero mean, unit variance). | COMPLETED |
| Step 3: Multi-Scale CNN Encoding | Dilated receptive fields (1x1, 3x3, 5x5) extracted hierarchical spatial context. | COMPLETED |
| Step 4: PennyLane Quantum PQC | Parameterized Ry, Rz quantum rotation gates evaluated on 4-qubit entangled state. | COMPLETED |
| Step 5: MC Dropout Uncertainty | Monte Carlo Bayesian sampling (T=10) estimated voxel-wise epistemic variance. | COMPLETED |
| Step 6: Entropy-Gated Attention | EGSA modulated skip-features with Laplacian edge supervision for sharp tumor borders. | COMPLETED |
| Step 7: Confidence Calibration | Calibrated confidence score derived strictly from model uncertainty map. | COMPLETED |

## 3. QUANTUM AI DIAGNOSTIC FINDINGS & METRICS

| Metric Parameter | Quantitative Finding | Clinical & Mathematical Interpretation |
|---|---|---|
| Tumor Detection Status | YES (Detected) | Enhancing Neoplasm Identified |
| Estimated Tumor Volume | 104 voxels (~1.04 cm2) | Axial 2D Cross-Section Coverage |
| Model Confidence Rate | 99.9% | Calibrated Strictly from Model Epistemic Variance |
| Mean Epistemic Uncertainty | 0.0 | MC Bayesian Boundary Variance Score |
| GPU Inference Latency | 1361.9 ms | Real-Time Acceleration (NVIDIA GeForce RTX 3050 6GB Lapt |

## 4. MULTI-MODAL SEGMENTATION & UNCERTAINTY HEATMAP VISUALS




** This clinical audit report was automatically generated by HQU-MSANet Quantum AI Architecture. Calibrated model confidence conforms to BraTS 2023 standards.*